

Higher Normal Forms

4NF, 5NF

Dependencies

- Functional Dependencies FDs: Role in decomposition and schema refinement
- Multivalued Dependencies
- Join Dependencies

4 NF

- BCNF removes any anomalies due to FDs
- Further research has led to the identification of another type of dependency called Multi-valued Dependency (MVD)
- Proposed by R Fagin* in 1977
- MVDs can also cause data redundancy
- MVDs are a generalization of FDs

** R Fagin Multi-valued Dependencies & a new normal form for relational databases, ACM TODS2, No. 3 (Sept. 1977)*

Multivalued Dependency

- Relation R (Course,Teacher,textbook) CTX
- Teacher T can teach course C and text X is recommended for this course
- No FDs exist
- Key is CTX
- Recommended texts for a course are independent of instructor
- CTX is in BCNF
- There is redundancy:
- Try inserting the fact that there is a new teacher for Physics101
- The text optics is a text for Physics 101 course is recorded once per potential teacher

MVD

- Consider the following relation CTX:

Course	Teacher	Texts
DBMS	M Bhise NDJ	RG Korth
MDA	M Bhise	G Booch McRobb

- In relational databases, repeating groups are not allowed

MVD

- 1 NF Version

CTX

<i>COURSE</i>	<i>TEACHER</i>	<i>TEXTS</i>
DBMS	M Bhise	RG
DBMS	M Bhise	Korth
DBMS	NDJ	RG
DBMS	NDJ	Korth
MDA	M Bhise	G Booch
MDA	M Bhise	McRobb

NO FDs in this relation

Anomalies

- New Teacher for DBMS
- New Text for DBMS
- Teacher teaching MDA leaves

Contd..

- MVD: Texts for a course are independent of the instructors
- CTX has no FDs at all (redundancies !!!)
- Use 2 binary relationship sets : Instructor (C,T) and Text (C, X) as these 2 are independent relationships
- If there is a tuple showing that C is taught by teacher T
- And there is a tuple showing that C has a book X as text
- Then there is tuple showing that C is taught by T and has text X
- If tuples (c,t1,x1), (c,t2,x2) appear then (c,t1,x2) and (c,t2,x1) also appear

MVD

- The relation CTX is not in 4NF as $C \twoheadrightarrow T$ is a nontrivial MVD and C is not a key
- But each of CT and CX are in BCNF
- If a relation is in BCNF, and at least one of its keys consist of single attribute then it is also in 4NF

4 NF

- Relation in BCNF and non-trivial MVDs absent

4 NF

- Decompose CTX into CT & CX

CT

<u>COURSE</u>	<u>TEACHER</u>
DBMS	M Bhise
DBMS	NDJ
MDA	M Bhise

CX

<u>COURSE</u>	<u>TEXT</u>
DBMS	RG
DBMS	Korth
MDA	G Booch
MDA	McRobb

4 NF

- Decompose CTX into CT and CX
- Decomposition of CTX into CT & TX is not done on the basis of FDs (as there are no FDs)
- Decompose CTX into CT & TX is done on the basis of MVDs
- MVDs

Represents a dependency between attributes of a relation, such that for every value of A, there is a set of values of B & a set of values of C, The set of values for B & C are independent of each other

course $\twoheadrightarrow$ *teacher* (course multi-determines teacher)

course $\twoheadrightarrow$ *text* (text multi-dependent on course)

MVD

- Decompose CTX into CT and CX
- Phy taught by a new teacher
- Nonlossy decomposition
- CTX is in BCNF (all key)
- CT and CX are also in BCNF (all key)
- MVDs are generalization of FDs
- Every FD is an MVD (but converse is not true)
- CTX has 2 MVDs $\text{course} \twoheadrightarrow \text{teacher}$, $\text{course} \twoheadrightarrow \text{Text}$ (teacher multidependent on course)
- Each course has well defined set of teachers and well defined set of texts

Contd..

- MVD Definition : Let A,B,C be subsets of relation R, we can write $A \twoheadrightarrow B$ if and only if, in every possible legal value of R, the set of B values matching a given (A value, C value) pair depends only on the A value and is independent of C value
- Fagin theorem tells you that MVDs go in pairs
- $A \twoheadrightarrow B$ holds only if MVD $A \twoheadrightarrow C$ also holds $A \twoheadrightarrow B \mid C$
- Every FD is an MVD in which the set of dependent (RHS) values matching a given determinant (LHS) value is always a singleton set

MVDs

- Some MVDs are not FDs
- The existence of such MVDs in CTX requires: to insert 2 tuples to add a new physics teacher
- These 2 tuples are needed to maintain an integrity constraint that is presented by $C \twoheadrightarrow X$
- CT and CX don't include any such MVDs
- A multi-valued dependency occurs when a determinant determines more than one dependent, and the dependents are independent of each other

Fagin Theorem

- Stronger version of Heath's Theorem
- Let $R\{A,B,C\}$ be a relation, where A,B,C are sets of attributes. Then R is equal to the join of its projections on $\{A,B\}$ and $\{A,C\}$ if and only if R satisfies the MVDs $A \twoheadrightarrow B|C$

4 NF

- An MVDs $A \twoheadrightarrow B$ is **trivial** if
 - (a) $B \subseteq A$ **or**
 - (b) $A \cup B = R$
- **A relation that is in BCNF & contains no non-trivial MVDs is said to be in 4NF**
- CTX is not in 4NF because *course* $\twoheadrightarrow$ *teacher* is a non trivial MVD
- What about CT and CX?

MVD

- $R(C,T,X)$ has no FDs and is all key, hence in BCNF
- Has redundancies
- MVDs $C \twoheadrightarrow T|X$
- $R_1(C,T)$ $R_2(C,X)$ have no FDs, in BCNF
- No redundancies
- $C \twoheadrightarrow T$ and $C \twoheadrightarrow X$ are trivial MVDs
- Although $T, X \subseteq C$, $C \cup T = R_1$ and $C \cup X = R_2$

Multi-Valued Dependencies

- Most common source of redundancy in BCNF schemas is to put 2 or more M:M relationships in a single relation

Fourth Normal Form 4NF

- R is in 4NF if and only if, whenever there exist subsets A and B of attributes of R such that the nontrivial MVD $A \twoheadrightarrow B$ is satisfied, then all attributes of R are also functionally dependent on A
- The only nontrivial dependency (FDs or MVDs) in R are of the form $K \rightarrow X$
- R is in 4NF if it is in BCNF and all (nontrivial) MVDs in R are in fact FDs out of keys
- CTX is not in 4NF since it involves MVD that is not an FD at all, let alone an FD out of a key
- MVD $A \twoheadrightarrow B$ is trivial if either A is a superset of B or the union of A and B is the entire R

Suppliers-Parts-Projects Database SPJ

S#	P#	J#
S1	P1	J2
S1	P2	J1
S2	P1	J1
S1	P1	J1

Join Dependencies

- n- decomposable relation
- (but not any m where $m < n$)
- SPJ (supplier#, part#, project#) is the join of SP, PJ and JS
 - If pair (s1,p1) appears in SP
 - And pair (p1,j1) appears in PJ
 - And the pair (j1,s1) appears in JS
 - Then the triple (s1,p1,j1) appears in SPJ

Contd...

- If s_1 is linked to p_1 , p_1 is linked to j_1 and j_1 is linked to s_1 , then s_1, p_1 and j_1 coexist in the same tuple
- A relation is n decomposable for some $n > 2$ if and only if it satisfies some such n -way cyclic constraint
- If (s_2, p_1, j_1) is inserted then (s_1, p_1, j_1) must also be inserted to validate JD integrity constraint

Join Dependency

- JD is a constraint on the set of legal relations over a database schema. An instance of relation R is subject to a join dependency if it can always be recreated by joining multiple tables each having a subset of the attributes of R
- If one of the tables in the join has all the attributes of the table T , the join dependency is called trivial

Join Dependency

Let R has subsets of attributes $A, B, \dots, Z$ then we say that R satisfies $JD \{A, B, \dots, Z\}$ if and only if every possible legal value of R is equal to the join of its projections on $A, B, \dots, Z$

$JD \{SP, PJ, JS\}$

Fagin's Theorem (modified)

$R\{A,B,C\}$ satisfies $JD^*\{AB,AC\}$ if and only if it satisfies the MVDs
 $A \twoheadrightarrow B|C$

MVD is a special case of JD (Like FD is a special case of MVD)

Nontrivial JD

- $JD^* \{A, B, \dots, Z\}$ is trivial if and only if one of the projections $A, B, \dots, Z$ is the identity projection of R (ie projection over all attributes of R)

JD and MVD

- 2-ary join dependencies are called multivalued dependency. More specifically if U is a set of attributes and R a relation over it, then R satisfies $X \twoheadrightarrow Y$

if R satisfies

$$*(X \cup Y, X \cup (U - Y))$$

5 NF

- $A \twoheadrightarrow B|C = \{AB, AC\}$
- Most general form of dependency possible
- SPJ involves a JD that is not an MVD (hence not FD either)
- **A relation is in 5NF (projection-join normal form PJ/NF) if and only if**
 - it is in 4 NF and
 - every nontrivial join dependency that holds for R is implied by the candidate keys of R
- SPJ can be decomposed into SP, PJ and JS and this is not implied by the fact that the combination $(S\#, p\#, j\#)$ is a candidate key
- But SP, PJ, JS are in 5NF each

5 NF

Projection-Join NF (PJ/NF)

- A relation is in 5NF if and only if every non-trivial join dependency in the relation is implied by the candidate keys.
- A join dependency $\{A, B, \dots, Z\}$ on R is implied by the candidate key(s) of R if each of A, B, ..., Z is a superkey for R.